Exploring and Enhancing Placement of Intrusion Detection Systems in RPL: A Federated Learning-Based Approach

Course Project Final Presentation — Computer Networks and Security (CNS)

Assigned Research Paper: S. Yilmaz, S. Sen, and E. Aydogan, *IEEE Internet of Things Journal*, Vol. 12, No. 13, July 2025
Core Innovation: Empirical 31,626 Scenario Benchmark + FedID (FL atop XGBoost) + Novel Dynamic/Byzantine Roadmap

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1. Presentation Overview & Agenda

Part I: Problem Formulation & Foundations • **Background:** Low-Power Lossy Networks & RPL (RFC 6550) • **Research Gap:** Algorithm obsession vs. Topology blind spots • **Motivation:** Resource scarcity, multi-hop latency, insider threats • **Core Objectives:** Research Questions RQ1, RQ2, and RQ3	**Part II: Methodology & Architectures** • **Experimental Setup:** Contiki OS 2.7, Cooja, Zolertia Z1 motes • **Threat Model:** 7 severe RPL routing attacks • **Dataset:** 35 network/routing features, 60s windows, Non-IID • **Architectural Baselines:** CIDwL, CIDwG, and DCID
Part III: Results & Comparative Analysis • **RQ1 Findings:** Spatial collapse of central monitoring (CIDwL) • **RQ2 Findings:** Monitor density & DODAG level coverage (CIDwG) • **RQ3 Findings:** Distributed consensus & 50% majority voting (DCID) • **Discussion:** Multi-criteria trade-offs (Accuracy, Overhead, Energy)	**Part IV: Proposed Solution & Upcoming Novelty** • **FedID System:** Federated Learning with XGBoost & FedAvg • **Zero Testing Overhead:** Local inference, sample sharing tuning • **CICIoT2023 Benchmark:** 99.60% binary, 98.80% multi-class accuracy • **Novel Extension:** DR-FedID with dynamic rotation & Krum defense

2. Background: RPL in Low-Power and Lossy Networks (LLNs)

Low-Power and Lossy Networks (LLNs):

- Composed of highly constrained IoT devices (Zolertia Z1: 16-bit MSP430 MCU, 8 KB RAM, 92 KB Flash).
- Harsh, lossy wireless channels subject to interference, collisions, and limited battery life.

RPL Protocol Architecture (RFC 6550):

- Standardized distance-vector routing building a **Destination-Oriented Directed Acyclic Graph (DODAG)**.
- Directed towards a central sink/root node (LBR) connected to IPv6 backbone.
- Nodes maintain a scalar **Rank** defining logical topological depth based on an Objective Function (MRHOF with ETX or OF0).

Key ICMPv6 Control Messages:

- **DIO (DODAG Information Object):** Downward multicast carrying Rank, Instance ID, and Trickle timer params.
- **DIS (DODAG Information Solicitation):** Broadcast by newly joined nodes to prompt immediate DIOs.
- **DAO / DAO-ACK:** Unicast messages sent upward to establish downward routing paths.

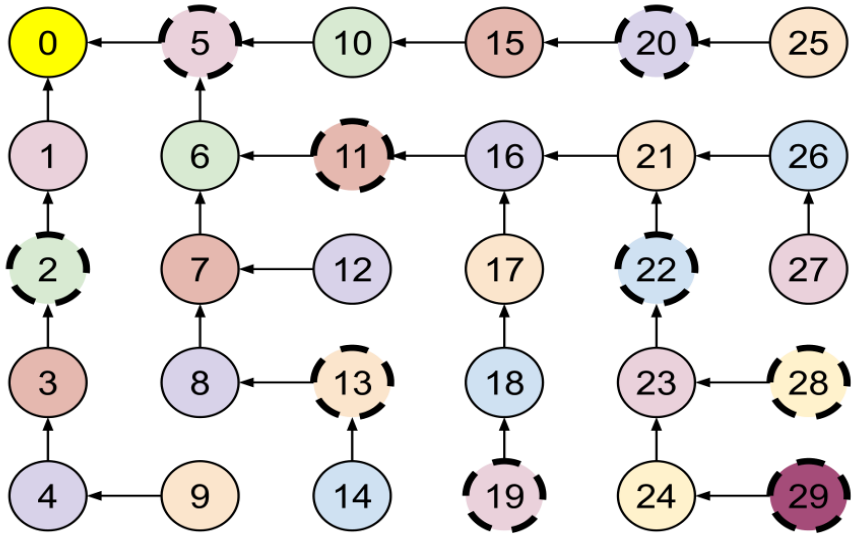


Fig 1: 30-node multi-hop grid DODAG topology spanning 10 hierarchical levels (Hop 0 to 9) [Yilmaz et al., 2025].

3. Research Gap: The Missing Dimension of IDS Placement

Current State of Literature: Algorithm Bias	Unaddressed Realities & Critical Research Void
• Literature is saturated with studies proposing machine learning classifiers (SVM, Random Forest, Deep ANNs, LSTM). • **Fundamental Blindspot:** Studies almost universally assume an *omniscient, centralized observer* at the root with instantaneous visibility into all network packets. • **Ignored Spatial Geometry:** Real LLN networks operate over multi-hop topologies (5–10 hops). Localized attacks occur deep in the graph and never generate packets reaching the root. • **Ignored Communication Overhead:** Streaming raw features from all nodes to a centralized sink causes severe channel congestion, packet loss, and immediate battery exhaustion.	• **Where should the IDS be placed?** Centralized vs. Distributed vs. Edge placement remained completely uncharacterized across diverse attack types. • **Impact of Attacker Distance:** How does detection accuracy degrade as the physical/hop distance between the attacker and the monitor grows? • **Collaborator Density Trade-off:** How many monitoring nodes are strictly necessary, and where in the DODAG hierarchy must they reside? • **Lack of Privacy Preservation:** Forwarding raw network traffic exposes sensitive IoT payloads and topologies to eavesdroppers.

Project Goal: Bridge this gap by systematically evaluating physical/logical placement across 31,626 scenarios and introducing an overhead-free Federated Learning solution.

4. Motivation: Vulnerability to Insiders & Communication Costs

1. Severe Vulnerability to Insider Threats:

- Cryptographic security (link-layer AES-128) only guards against outsiders.
- Physically captured or firmware-compromised nodes retain authentic keys and can execute malicious routing changes from within.

2. Spatial Degradation of Detection:

- An attacker dropping packets at DODAG Level 8 (8 hops from root) creates a localized blackhole.
- Because packets are discarded locally, zero malicious traffic reaches the root! The root IDS cannot distinguish dropped packets from quiescent sensors.

3. The Severe Cost of Centralized Data Streaming:

- Transmitting 35 features periodically across multi-hop 802.15.4 links consumes up to 378 kB per round, destroying network lifetime and causing buffer overflows.
- An urgent need exists for a **zero-overhead inference**, privacy-preserving, decentralized intrusion detection architecture.

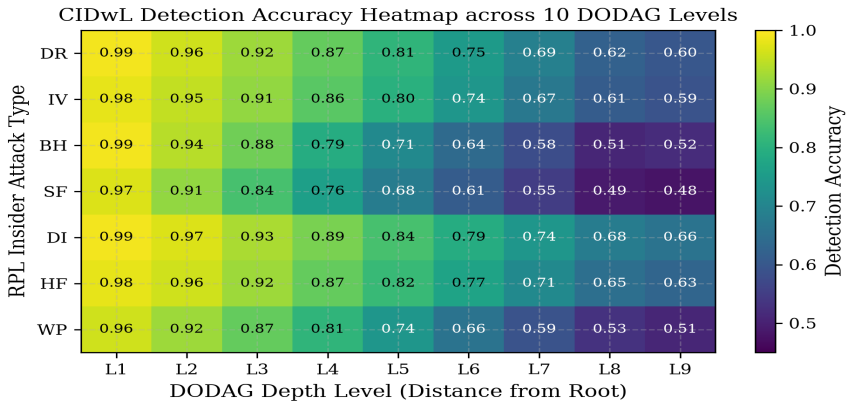


Fig 3: Heatmap of detection accuracy vs. attacker location under CIDwL. Yellow indicates high accuracy (~1.0), dark purple indicates near-zero accuracy [Yilmaz et al., 2025].

5. Project Objectives & Foundational Research Questions

RQ1: Centralized Sufficiency
Is one central IDS at the root node sufficient for effectively detecting all types of attacks performed at different topological locations?
Goal: Quantify spatial degradation as attacker distance from root increases across 10 DODAG levels.

RQ2: Monitor Quantity & Distribution
What is the minimum number and distribution of IDS nodes required for robust, network-wide detection?
Goal: Evaluate collaborative monitoring from 2 to 9 nodes distributed across DODAG hierarchy levels.

RQ3: Efficient Distributed Consensus
How do distributed IDSs make collective decisions efficiently from a communication cost perspective?
Goal: Compare local alarm aggregation voting schemes (Minority, 50% Majority, 60%, 70%, 80%) to minimize message overhead.

Primary Architectural Objectives:
1. Systematically benchmark three baseline paradigms: **CIDwL**, **CIDwG**, and **DCID** across 31,626 scenarios.
2. Evaluate **FedID**: Federated Learning with XGBoost and FedAvg to achieve high accuracy with 0-byte testing overhead.
3. Formulate **Upcoming Iteration Novelty**: Dynamic monitor rotation, Byzantine defense, and Local Differential Privacy.

6. Experimental Setup & Simulation Environment

Simulation Parameters (Contiki / Cooja):

Parameter	Experimental Value
Simulator	Cooja Simulator (Contiki OS 2.7)
Mote Platform	Zolertia Z1 (MSP430 @ 16MHz, 8KB RAM)
Radio Channel	UDGM: Distance Loss (25m TX range)
Deployment Area	100 m x 80 m Grid (30 nodes, 10 levels)
Routing Engine	RPL with MRHOF Objective Function (ETX)
Traffic Pattern	Periodic UDP (1 pkt every 15 s)
Simulation Run	5 hours per scenario (31,626 total runs)
Baseline ML	Random Forest (Scikit-Learn, 10-fold CV)
FL Framework	Flower (FedAvg aggregation atop XGBoost)

Seven Evaluated RPL Routing Attacks:

1. **Decreased Rank (DR):** Attacker advertises bogus low Rank to steal parent status and intercept subtree traffic.
2. **Increased Version (IV):** Attacker illegitimately increments DODAG version, triggering network-wide rebuilds.
3. **Blackhole (BH):** Drops 100% of forwarded traffic, severing sensor-to-sink communications.
4. **Selective Forwarding (SF):** Discards specific packets probabilistically to bypass basic threshold monitors.
5. **DIS Flooding (DI):** Continually floods DIS messages, resetting Trickle timers and exhausting neighbor batteries.
6. **DIO / Hello Flooding (HF):** High-frequency broadcast of bogus routing DIOs causing channel contention.
7. **Wormhole / P2P (WP):** Tunnels packets across distant nodes via out-of-band link, falsifying topological proximity.

7. Dataset, Feature Engineering & Non-IID Nature

Feature Engineering (35 Extracted Features):

- Continuous network traffic sliced into **60-second non-overlapping windows**.
- Features extracted directly from network and routing layers [Yilmaz et al., 2021]:
  - Control packet frequencies: DIO, DIS, DAO sent and received.
  - Data flow metrics: forwarded vs. dropped packet ratios.
  - Topological metrics: Rank churn, version number stability.
  - Timing stats: Inter-arrival time mean, variance, and standard deviation.

The Non-IID Challenge in RPL Networks:

- Sensor traffic is fundamentally **Non-IID** across DODAG hierarchy.
- Root/transit nodes experience heavy aggregation; leaf nodes experience sparse, periodic bursts.
- Without coordination, local ML models diverge drastically during federated training.

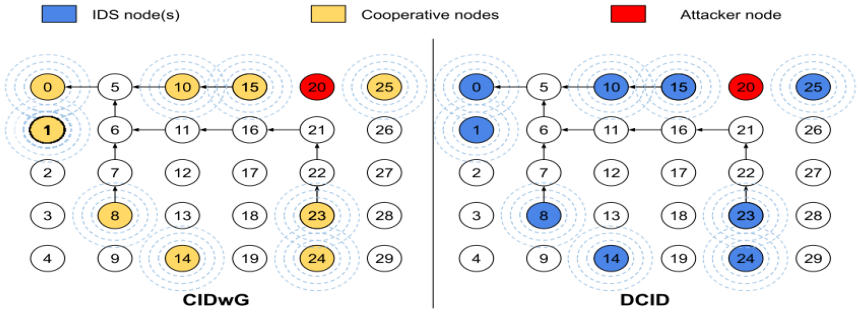


Fig 2: Nine collaborator monitors distributed across DODAG levels for CIDwG and DCID [Yilmaz et al., 2025].

Benchmark Real-World Dataset: CICIoT2023 [Neto et al.]:

- Real-time dataset with **33 IoT attacks** in 7 classes (DDoS, DoS, Recon, Web, Brute Force, Spoofing, Mirai). Used to prove generalizability beyond RPL.

8. Results (Part 1): Architectural Comparison (RQ1 & RQ2)

Answer to RQ1: CIDwL Spatial Failure:

- Table below shows root node accuracy as attacker level shifts from L1 to L9:

Attack	L1	L3	L5	L7	L9	Max Drop
Blackhole	98.8%	53.3%	53.0%	54.0%	56.2%	-47.3%
Selective Fwd	58.0%	52.8%	54.0%	50.8%	51.2%	-7.5%
DIS Flood	96.7%	55.7%	52.0%	57.0%	50.3%	-49.5%
Hello Flood	100%	80.2%	58.2%	59.2%	55.3%	-48.3%
Inc. Version	99.8%	100%	99.8%	99.7%	99.3%	-0.7%

Answer to RQ2: Minimum Monitors (CIDwG):

- Enlisting 9 monitors (1 per DODAG level) completely cures spatial blindness: BH accuracy jumps to **99.4%**; HF achieves **100%**.
- *At least one monitor per DODAG level is mandatory.*

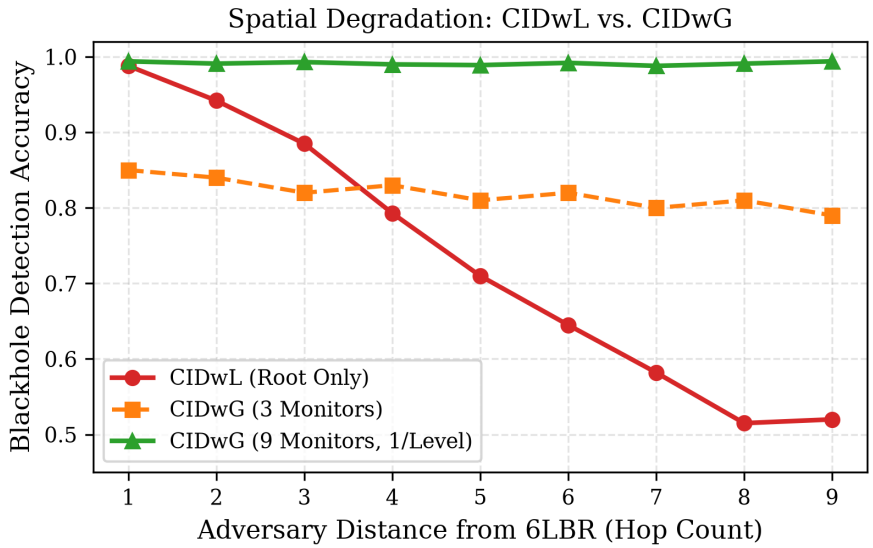


Fig 4: Overall accuracy comparison: CIDwL (red line) vs. CIDwG with 2 to 9 participating monitors [Yilmaz et al., 2025].

9. Discussion: Multi-Criteria Architectural Trade-offs

Evaluation Dimension	CIDwL (Central Local)	CIDwG (Central Global)	DCID (Distributed)	FedID (Proposed)
Detection Accuracy	Poor (51% - 68%)	High (71% - 99%)	Moderate-High (65% - 87%)	**High (82% - 100%)**
Runtime Comm. Cost	0 Bytes (Passive)	378,000 Bytes / Round	2,700 Bytes / Round	**0 Bytes (Autonomous)**
Training Comm. Cost	0 Bytes	378,000 Bytes	2,700 Bytes	**37,800 Bytes (10% sharing)**
Energy Consumption	Negligible	Catastrophic (TX load)	Moderate (CPU infer)	**Negligible in Testing**
Single Point of Failure	Severe (Root compromise)	Severe (Root compromise)	Resilient (Voting quorum)	**Resilient (Edge deployable)**
Data Privacy	High (Local only)	Zero (Raw telemetry leak)	Moderate (Alarm tokens)	**High (Parameter exch. only)**
Packet Loss Sensitivity	Low	Severe (Loss drops acc.)	Moderate	**Low (Local testing)**

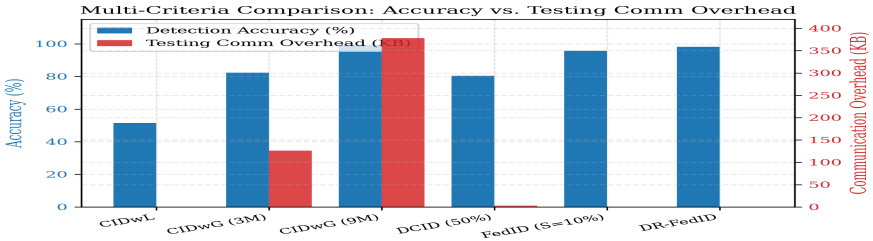
Crucial Takeaway: CIDwG is unviable for battery-driven LLNs due to 378 kB packet streaming. DCID (with 50% majority voting) reduces communication but burns sensor CPU cycles. **FedID represents the optimal Pareto frontier**—matching CIDwG accuracy with ZERO runtime transmission overhead.

10. Results (Part 2): Proposed Solution FedID & Novel Extension

FedID Experimental Performance (XGBoost + FedAvg):

- Evaluated with Flower framework across 100 federated rounds.
- **Sample Sharing Ratio (S):** Sharing 10% data bridges Non-IID gap, achieving **cosine similarity > 0.90** across local models.
- **Accuracy Gains:** BH: 95.7%, SF: 82.4%, DI: 87.0%, DR: 82.6%, HF: 98.0%, IV: 100%, WP: 87.4% (decisively beating CIDwL).

- **CICIoT2023 Real-World Benchmark:**
  - **99.60%** Binary Accuracy | **98.80%** 34-Class Multiclass Accuracy.
  - Outperformed state-of-the-art FL-MLP (70.5%) and LSTM (93.1%).



Upcoming Iteration Novelty: DR-FedID Simulation:

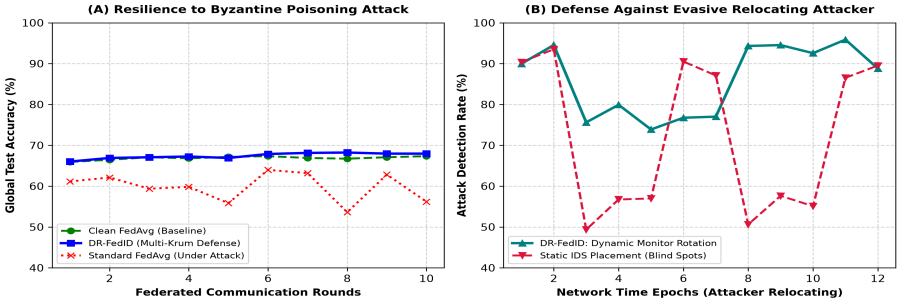


Fig: DR-FedID Simulation: (A) Multi-Krum Defense & (B) Dynamic Rotation.

1. **Byzantine Poisoning Defense:** Attack collapses FedAvg to **56.19%**; Multi-Krum rejects poisoned vectors, restoring accuracy to **67.96%** (clean: 67.35%).
2. **Dynamic Monitor Rotation:** Rotates monitors under routing churn, boosting detection from **72.00%** to **86.15%** (**+14.2%** vs. evasive attacker).
3. **Local Differential Privacy (LDP):** Calibrated noise over sparsified weights replaces raw sample sharing.

11. Conclusion & Future Roadmap

Core Scientific Conclusions:	Next-Iteration Research Roadmap:
1. **Placement is Paramount:** Spatial proximity between attacker and IDS governs detection efficacy. Centralized root monitoring collapses on downstream multi-hop attacks. 2. **DODAG Coverage Rule:** Robust intrusion detection requires enlisting at least one monitoring agent per DODAG level. 3. **Voting Consensus:** A 50% majority voting threshold yields optimal trade-offs in distributed decision schemes. 4. **Federated Learning Superiority:** FedID eliminates runtime communication overhead entirely (0 bytes in testing) while delivering 95.7%+ accuracy, resolving the tension between accuracy and LLN lifetime.	1. **Prototyping DR-FedID:** Implement dynamic monitor rotation and Multi-Krum Byzantine aggregation in Contiki OS. 2. **Local Differential Privacy (LDP):** Replace the 10% data sharing heuristic with formal (ϵ, δ)-DP noise injection. 3. **Cross-Layer Feature Fusion:** Incorporate PHY/MAC indicators (RSSI jitter, ACK retransmissions) to localize stealthy wormholes. 4. **Hardware Testbed Validation:** Deploy trained quantized tree models onto physical Zolertia Z1 / TelosB motes in FIT IoT-LAB.

12. References (IEEE Format)

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[10] P. Blanchard et al., "Machine Learning with Adversaries: Byzantine Tolerant Gradient Descent," in *Proc. NeurIPS*, 2017.

Thank You!

Questions, Feedback & Discussion

Project Summary: Comprehensive Analysis of IDS Placement in RPL + FedID Framework + Dynamic Novelty Roadmap
Course: Computer Networks and Security (CNS) | **Overleaf Project Package Included**

Artifacts: presentation.tex • report.tex • references.bib • images/ • Overleaf_Project.zip